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DOCROUTE-RT: A Real-Time AI-Based System for Document Monitoring, Sharing, and Routing

A Secure Framework for Intelligent Routing, Tracking, and Real-Time Analytics

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Abstract: Organizations are encountering a lot of difficulties with the use of technology to securely manage, share, and audit documents throughout their lifecycle while keeping transparency and operational efficiency. A majority of non-digital/former Document Management Systems (DMS) do not have real-time tracking abilities, workflow routing based on advanced technology, or detailed audit logs [1], [3]. This paper presents an AI-based system called DOCROUTE-RT to monitor, share, and route documents along with integrated real-time analytics reporting. The framework will include the use of Optical Character Recognition (OCR) to digitize documents and artificial intelligence methods to automatically summarize and categorize documents within the system [4], [5], [12]. A hybrid workflow model will utilize automated intelligence as well as human judgment to enable deadlines for routing, as well as reminders for due dates [6], [10]. Secure access to documents will be accomplished by utilizing QR code based sharing and code-based document exchange, as well as through the use of version control and detailed audit logs, enabling complete traceability of document related activities [7], [9]. Lastly, through the use of real-time dashboards, users will receive continuous monitoring of document related activities, system health, and user interaction, enabling accountability for all document related activities [1], [11], [15]. The use of this new system will increase the security, accountability, and efficiency of document lifecycle management in academic, corporate, and enterprise settings.

Index Terms - *Document Monitoring, Secure File Sharing, Hybrid Workflow, Artificial Intelligence, Optical Character Recognition (OCR), Document Tracking, QR Code Security, Workflow Routing, Real-Time Analytics, Audit Logging*

1. INTRODUCTION

The rapid growth of digital documentation has transformed how information is created, shared, and managed across academic, corporate, and enterprise environments [1], [2]. However, many existing document management systems provide limited support for real-time monitoring, secure sharing, intelligent routing, and comprehensive traceability [3], [6]. Documents typically pass through multiple stages such as creation, review, approval, and distribution, where insufficient visibility into access history, version changes, and deadlines can lead to unauthorized usage, compliance risks, and operational inefficiencies [3], [7]. Additionally, the continued reliance on physical and scanned documents highlights the need for effective digitization mechanisms that can seamlessly integrate with modern digital workflows [4], [5].

Recent advancements in Artificial Intelligence (AI), including Optical Character Recognition (OCR), automated summarization, and semantic tagging, have enabled intelligent document understanding and content extraction [4], [12]. Nevertheless, in most existing platforms these capabilities function as isolated features and lack integration with real-time workflow management, access control, and tracking mechanisms, underscoring the need for a balanced hybrid workflow approach [2], [6], [10].

To address these challenges, this paper presents DOCROUTE-RT, an AI-based system for real-time document monitoring, secure sharing, and workflow routing. The proposed unified framework integrates OCR-based digitization, AI-driven document summarization and classification, secure QR- and code-based document sharing, version control, and comprehensive audit logging [4], [5], [7], [9]. A hybrid workflow model combines automated intelligence with human decision-making to support deadline-aware routing and reminder management [6], [10], [15]. Furthermore, a real-time analytics dashboard provides continuous insights into document activity, workflow status, and system performance [1], [11], [15]. By combining intelligent automation, secure access control, and real-time analytics, DOCROUTE-RT enhances security, transparency, accountability, and efficiency across the entire document lifecycle.

2. LITERATURE REVIEW

2.1 Existing Tools and Methodologies

Existing document management systems (DMS) have evolved from basic scanning and storage solutions into platforms that incorporate Optical Character Recognition (OCR), Natural Language Processing (NLP), and rule-based workflow automation to improve operational efficiency and regulatory compliance [4], [12]. These systems typically support document digitization, metadata extraction, version control, role-based access control, and predefined workflow

processes [3], [7]; however, most rely on static, rule-driven mechanisms and lack advanced AI-based intelligence for real-time, content-aware decision-making. As a result, they provide limited support for dynamic workflow routing, continuous document tracking, and integrated analytics [2], [6]. In addition, document processing, monitoring, and reporting are often implemented as separate or loosely integrated components, leading to fragmented data, reduced transparency, delayed responses to workflow bottlenecks, and the absence of a unified, real-time view of the complete document lifecycle [1], [11].

2.2 Related Research Papers

Existing research has extensively explored OCR, NLP, and machine learning techniques for document understanding, extraction, and classification [4], [5], [12], [13], and several studies emphasize automated metadata extraction and summarization to improve retrieval and processing efficiency [2], [8]. Other works investigate machine learning–based workflow optimization and task routing to enhance operational performance [6], [10], [15], while event-driven and real-time analytics approaches have been proposed to support monitoring and decision-making in enterprise workflows [1], [6], [11]. However, many of these approaches treat document analysis and workflow management as separate problems, operating in offline or batch modes and lacking integrated real-time monitoring, secure physical document tracking, version-level auditability, and unified analytics dashboards [3], [7], [9], [14]. In addition, several routing models do not fully leverage document content, access patterns, or security constraints when making routing decisions [10], [15]. These limitations point to a clear research gap for a unified, real-time framework that combines intelligent document understanding, secure sharing and tracking, hybrid workflow routing, and live analytics—capabilities that DOCROUTE-RT aims to provide.

3. Problem Statement

Modern organizations manage a rapidly increasing volume of physical and digital documents that must be securely processed, shared, routed, and monitored across multiple departments and stakeholders [2], [3]. Existing document management systems primarily focus on digitization, storage, and basic rule-based workflows, offering limited support for intelligent content-aware routing, real-time document movement monitoring, secure physical document tracking, and actionable analytical insights [3], [6]. The lack of integration between document processing, workflow automation, tracking, and analytics results in reduced transparency, delayed processing, limited accountability, and difficulty in proactively identifying workflow bottlenecks [1], [11], [15]. Consequently, there is a critical need for a unified, real-time, and intelligent document management framework that enables end-to-end lifecycle visibility, secure tracking of both digital and physical documents, and data-driven workflow optimization—gaps that the proposed DOCROUTE-RT system aims to address [2], [10], [15].

4. Objectives of the Research

The primary objective of this research is to design and develop DOCROUTE-RT, a real-time AI-based system for document monitoring, sharing, and workflow routing that supports complete end-to-end document lifecycle management [2], [6]. The study aims to integrate OCR and NLP techniques for accurate document digitization, content extraction, and metadata generation, enabling efficient handling of both physical and digital documents [4], [5], [12]. Another objective is to employ AI-driven models within a hybrid workflow framework to support intelligent, content-aware routing, deadline management, and reminder automation while retaining human oversight [6], [10], [15]. The research also seeks to enable secure document sharing and real-time tracking through QR-based and code-based mechanisms, ensuring traceability, version control, and accountability [7], [9]. Finally, the study aims to provide real-time analytics dashboards that visualize document flow, processing delays, workflow status, and system health, and to evaluate the effectiveness of the proposed system in improving routing accuracy, reducing document turnaround time, and enhancing operational transparency compared to conventional document management systems [1], [11], [15].

5. Proposed Methodology and System Architecture

The intended system architecture and proposed methodology for DOCROUTE-RT, a real-time AI-powered document monitoring, sharing and routing workflow solution, are presented in this section. The methodology gives insight into how a document moves through the operational flow as well as how intelligent processing logic is applied to it. The system architecture provides an overview of the layered structural design that will provide secure, scalable, end-to-end management of all types of documents throughout their lifecycle whether physical or digital.

5.1 Overall Methodology

The overall methodology of DOCROUTE-RT follows an intelligent, event-driven document lifecycle approach in which documents are ingested from multiple sources and digitized using Optical Character Recognition (OCR) techniques. The extracted content is processed using Natural Language Processing (NLP) and machine learning models to identify metadata, classify document types, generate summaries, and detect content modifications. Based on this extracted intelligence, documents are automatically routed through optimized workflows determined by document priority, user roles, and departmental workload within a hybrid workflow framework that combines automated decision-making with human oversight. Throughout the document lifecycle, both digital and physical documents are continuously tracked using unique identifiers and secure QR codes, while all access events, workflow actions, approvals, and modifications are logged in real time. The system continuously feeds operational data into a live analytics engine, enabling real-time visualization of document movement, processing delays, and workflow performance, thereby ensuring automation, transparency, accountability, and proactive workflow optimization.

5.2 System Architecture

The DOCROUTE-RT system employs a modular, layered architecture that enables users to monitor documents in real-time, securely share them possessively, intelligently route them for workflow processing, and make analytical decisions. This architecture allows the seamless integration of digital and physical document workflows while providing scalability, security, and extendibility.

Each layer has a specific function and communicates with adjacent layers via secure, well-defined interfaces, thereby facilitating end-to-end management of documents throughout their complete life cycle.

The architecture of DOCROUTE-RT consists of six functionally separate layers. Figure 5.2 illustrates the overall system architecture of DOCROUTE-RT, showing the interaction between document ingestion and OCR-based digitization, AI-driven content processing, real-time document

tracking, intelligent hybrid workflow routing, secure access control with audit logging, and real-time analytics for monitoring document flow and workflow performance.

Figure 5.2: DocRoute-RT System Architecture

5.3 System Components

Table 1 presents the key components of the proposed DOCROUTE-RT system.

❖ **Table 1: System Component: -**

Component	Description
Document Ingestion & Digitization	Handles document intake from multiple sources and applies OCR to convert documents into machine-readable text while extracting metadata.
AI Intelligence & Content Processing	Uses NLP and machine learning techniques for document classification, summarization, entity extraction, and integrity verification.
Document Tracking (Digital & Physical)	Provides real-time monitoring of documents using unique tracking identifiers and encrypted QR codes.
AI-Based Workflow & Lifecycle Management	Automates content-aware routing, deadline assignment, and anomaly detection using a hybrid workflow approach.
Security, Access Control & Audit	Enforces role-based permissions, secure sharing, approval mechanisms, and maintains immutable audit logs.
Real-Time Analytics & Visualization	Aggregates system data and presents live dashboards for monitoring document flow and workflow performance.

5.4 Workflow Diagram

The workflow diagram, as shown in Figure 5.4, illustrates the end-to-end operational flow of documents in the DOCROUTE-RT system. This is an example of how a variety of documents are gathered from different sources, processed using Optical Character Recognition (OCR) and Artificial Intelligence (AI)-based content analysis to produce text, metadata and context data from the documents. Once the data is collected, decision will be made on how to route this data through a Hybrid Workflow Framework based on the data that was collected, using content-based criteria and prioritised criteria to make content-aware and human-supervised decisions about how to route the documents to the appropriate users. The documents, both digitally and physically, will be uniquely identified and trackable throughout the workflow by assigning them a unique identifier and a secure QR code, and all actions taken in the workflow and whenever the documents are accessed or modified will be captured in real-time logs while analytics and monitoring components provide live insights into workflow performance, document movement, and processing delays, enabling continuous monitoring and optimization of document workflows. In addition, automated alerts and deadline management mechanisms ensure timely task execution and escalation of delays, and feedback loops between workflow execution and analytics modules allow the system to adapt routing strategies dynamically resulting in a closed-loop design that enhances operational efficiency, accountability, and overall system responsiveness.

Figure 5.4: DocRoute-RT Workflow

5.5 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD), as shown in Figure 5.5, illustrates the movement of data across the core components of the DOCROUTE-RT system. It represents how document data flows from users and departments through document ingestion and digitization, AI-based content processing, intelligent workflow routing, and document tracking modules. The DFD highlights interactions between processing units, data stores, and external entities, including the document repository, metadata and classification store, and audit and log store. Furthermore, workflow actions, tracking updates, and access events are synchronized in real time and forwarded to the analytics module, where document flow, workflow performance, and system metrics are visualized through live dashboards, ensuring transparency and end-to-end document lifecycle management.

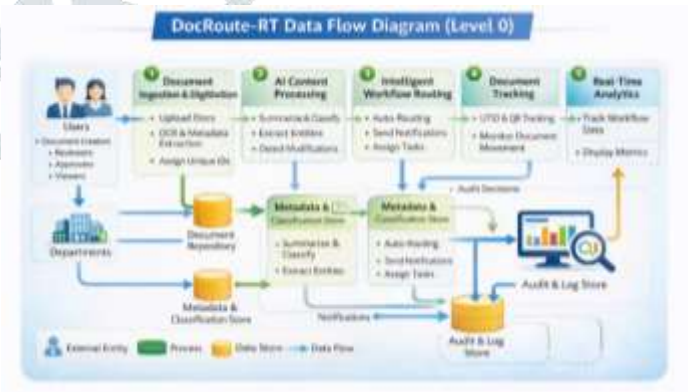

Figure 5.5: DocRoute-RT (DFD Diagram)

6. IMPLEMENTATION DETAILS

The DOCROUTE-RT system is implemented using modern, scalable technologies to support real-time document monitoring, secure sharing, intelligent workflow routing, and analytics. Python is used as the core development language due to its strong support for AI and data processing, Tesseract OCR provides the ability to accurately extract text from scanned documents/images. The use of NLP libraries like spaCy and NLTK for metadata extraction, entity recognition, and summarizing text; combined with the implementation of machine learning models (Random Forest/Transformer) for

classifying documents and creating content-aware workflows, will assist in establishing secure tracking mechanisms by generating encrypted QR codes then scanning those codes in real time to monitor physical and electronic documents. Document storage, document metadata management, version control, and document audit logs all use relational databases, complemented by NoSQL or log-based storage for high-frequency tracking and analytics data. Real-time dashboards are developed using web-based visualization libraries to present workflow performance and system health, while automated notifications via email and messaging services ensure timely alerts and effective end-to-end document lifecycle management. The modular implementation allows independent scaling of AI, workflow, and analytics components, improving system reliability. Security best practices such as encrypted communication and access validation are applied to protect sensitive document data.

7. Experimental Evaluation

This section presents a comprehensive experimental evaluation of DOCROUTE-RT to assess its functional correctness, performance, and reliability under real-world–like scenarios. The evaluation focuses on validating the system's ability to digitize documents, analyze content using AI, route documents through hybrid workflows, enforce access control during sharing, securely track both digital and physical documents, and provide real-time analytical insights. The experiments were designed to reflect the system's intended access model, consisting of a single end user (owner) and public users with restricted permissions. [2], [6], [15].

7.1 Evaluation Setup

The experimental setup was conducted in a controlled environment simulating an organizational document workflow managed by a single end user. A dataset comprising scanned images, PDF files, and DOCX documents with varying sizes, languages, and content complexity was used. The end user was responsible for document creation, editing, workflow routing, sharing, and access revocation. Public users accessed documents only through sharing mechanisms such as QR codes or secure access codes. The system was tested under normal operating conditions and moderate concurrent access to evaluate real-time tracking, audit logging, and analytics responsiveness [2], [6], [15].

7.2 Evaluation Metrics

The system was evaluated using the following key metrics[1], [2], [6], [12], [15]:

- OCR Accuracy: Correctness of text extraction from scanned and image-based documents [4], [5], [12]
- AI Processing Quality: Relevance and consistency of summaries, tags, and classifications [2], [8], [13]
- Workflow Routing Efficiency: Correct routing based on priority, role, and document type [6], [10], [15]
- Document Turnaround Time: Time taken from ingestion to workflow completion [6], [15]
- Tracking & Audit Reliability: Accuracy of access logs and version history [3], [7], [9]
- Access Control Enforcement: Correct restriction of public users during document sharing [7], [9].
- Analytics Responsiveness: Latency in updating real-time dashboards [1], [6], [11]

7.3 Test Cases

The table below provides Test Cases : -

❖ **Table 2: Test Cases Table: -**

Test Case ID	Test Description	User Type	Input / Action	Expected Outcome
TC-01	OCR Digitization	End User	Upload scanned image	Text extracted and stored in history
TC-02	OCR on Low-Quality Scan	End User	Upload blurred image	Partial extraction with minor errors
TC-03	AI Summarization	End User	Upload PDF/DOCX	Summary generated as per selected options
TC-04	Document Classification	End User	Upload mixed documents	Correct document type classification
TC-05	Workflow Routing	End User	Assign priority & deadline	Routed to appropriate workflow stage
TC-06	Hybrid Workflow Override	End User	Manual approval/edit	Human intervention allowed
TC-07	Digital Document Sharing	End User → Public	Share with defined permissions	Access limited to assigned permissions
TC-08	QR-Based Physical Tracking	Public User	Scan QR code	View-only access granted
TC-09	Unauthorized Modification Attempt	Public User	Attempt edit/download via QR	Access denied and logged
TC-10	Version Control	End User	Edit document	New version created and tracked
TC-11	Audit Logging	System	View/Edit/Download action	Immutable audit entry generated

Test Case ID	Test Description	User Type	Input / Action	Expected Outcome
TC-12	Secure File Exchange	End User → Public	Code-based access	File accessible only with valid code
TC-13	Analytics Update	System	Workflow/tracking event	Dashboard updated with minimal delay

7.4 Results and Observations

The experiments indicate that, with regard to the components tested, DOCROUTE-RT provides adequate performance throughout. The OCR produced a high level of accuracy on clean documents and lower levels of accuracy on low-quality scans. This, of course, is consistent with previous findings regarding the inherent limitations of OCR. The AI-based features for summarizing, tagging, and classifying documents allowed for efficient routing of workflows based upon the content, hence very little manual intervention was required to process documents or to route them to the correct location while supporting hybrid workflows. The QR-based tracking provided reliable logging of access events along with view-only restrictions being placed upon public users of the system. This provided security and traceability of the end-user access to the documents. The analytics dashboard was updated in near real-time, thereby confirming that the DOCROUTE-RT solution is capable of providing live monitoring of performance and assisting with better decision-making.

7.5 Discussion

The evaluation confirms that DOCROUTE-RT successfully integrates document digitization, AI-driven intelligence, secure sharing, access control enforcement, real-time tracking, and workflow automation within a unified framework. Compared to traditional document management systems, DOCROUTE-RT offers improved transparency, faster routing, and stronger accountability. The explicit separation between end-user control and public-user access enhances system security, particularly in physical document tracking scenarios.

7.6 Summary of Evaluation

The evaluation conducted proved that DOCROUTE-RT provides an intelligent, secure real-time solution for monitoring documents and routing workflow, therefore meeting the goals of this research. In addition, we validated its ability to improve operational efficiency, provide enhanced visibility of documents, place access controls on documents and assist with decision making. Lastly, we were able to identify continued areas for improvement such as; more advanced OCR models and scalability.

8. Security Analysis

Security in DOCROUTE-RT is ensured through a strict access control model that separates a single end user (owner) from public users. The end user retains full control over document management and sharing, while public users access documents only through authorized mechanisms, with QR-based physical tracking limited to view-only access. Role-based access control, secure sharing links, and immutable audit logs enforce accountability and prevent unauthorized actions. Encrypted tracking identifiers, version control, and secure communication channels further protect data integrity across both digital and physical document workflows.

9. CONCLUSION

This paper presented DOCROUTE-RT, a real-time AI-based system for document monitoring, sharing, and workflow routing designed to address the limitations of conventional document management systems. By integrating OCR-based digitization, AI-driven document analysis, hybrid workflow automation, secure tracking mechanisms, and real-time analytics, the proposed system enables efficient, transparent, and secure document lifecycle management.

Experimental evaluation demonstrates that DOCROUTE-RT improves workflow routing efficiency, enforces strict access control, and provides reliable real-time monitoring and auditability for both digital and physical documents. The hybrid workflow model effectively balances automated intelligence with human decision-making, ensuring flexibility while maintaining control and accountability. Overall, DOCROUTE-RT offers a scalable and practical solution for academic, corporate, and enterprise environments requiring intelligent, secure, and real-time document management.

10. Future Work

Future enhancements to DOCROUTE-RT will focus on improving OCR accuracy through advanced image pre-processing and deep learning–based OCR models, particularly for low-quality and handwritten documents. The system can be extended with more sophisticated AI models for context-aware workflow optimization and predictive analytics to further reduce processing delays. Additional security features such as blockchain-based audit logging and fine-grained access policies may be explored to strengthen trust and compliance. Finally, scalability improvements and large-scale deployment studies will be conducted to support enterprise-level usage and multi-organization document workflows.

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